

# DevOps Curriculum — Tools & Concepts

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# 1 DevOps & AWS Learning Hub

A structured path from zero to advanced, plus two handbook series:

| Resource                          | Target size    | Status                                          |
|-----------------------------------|----------------|-------------------------------------------------|
| <a href="#">DevOps Curriculum</a> | Reference list | Complete                                        |
| <a href="#">AWS Curriculum</a>    | Reference list | Complete                                        |
| <a href="#">DevOps Handbook</a>   | ~1,200 pages   | TOC + Part I-II written; Parts III-XII outlined |
| <a href="#">AWS Handbook</a>      | ~800 pages     | TOC + Part I-II written; Parts III-X outlined   |

## 1.1 Download PDFs

Pre-built PDFs are in the [pdf/](#) folder:

| PDF                                           | Description                              |
|-----------------------------------------------|------------------------------------------|
| <a href="#">devops-handbook.pdf</a>           | DevOps handbook — 11 chapters + full TOC |
| <a href="#">aws-handbook.pdf</a>              | AWS handbook — 11 chapters + full TOC    |
| <a href="#">devops-curriculum.pdf</a>         | DevOps tools & concepts reference        |
| <a href="#">aws-curriculum.pdf</a>            | AWS services reference                   |
| <a href="#">devops-aws-complete-guide.pdf</a> | All content in one PDF                   |

**Rebuild PDFs** after editing markdown:

```
./scripts/build-pdfs.sh
```

Requires `pandoc` and `wkhtmltopdf`.

1. **Start with the curricula** — they list every tool, concept, and AWS service in learning order.
2. **Follow the handbook TOCs** — each chapter has a page budget; work through sequentially.
3. **Expand on demand** — ask an AI assistant or instructor to flesh out any chapter number from the TOC.

## 1.2 Suggested study pace

| Phase        | Duration (self-paced) | Focus                                |
|--------------|-----------------------|--------------------------------------|
| Foundations  | 4-8 weeks             | Linux, Git, networking, scripting    |
| Core DevOps  | 8-12 weeks            | CI/CD, containers, IaC, cloud basics |
| Intermediate | 8-12 weeks            | Kubernetes, observability, security  |
|              |                       | CSP platform                         |

|          |           |                                                |
|----------|-----------|------------------------------------------------|
| Advanced | 12+ weeks | SRE, platform engineering, multi-cloud, FinOps |
|----------|-----------|------------------------------------------------|

## 1.3 Page convention

Each handbook “page”  $\approx$  400 words of instructional content + diagrams/commands. Labs and exercises add extra pages in the TOC estimates.

# 2 DevOps Curriculum: Tools & Concepts (Beginner → Advanced)

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Complete reference of what to learn, in recommended order. Items marked **(adv)** are advanced topics.

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## 2.1 Level 0 — Prerequisites (before DevOps)

### 2.1.1 Concepts

- Software development lifecycle (SDLC)
- Agile, Scrum, Kanban
- Waterfall vs iterative delivery
- Client-server architecture
- APIs (REST, GraphQL basics)
- Databases (SQL vs NoSQL overview)
- Environment tiers: dev, staging, production

### 2.1.2 Tools & Skills

| Category          | Tools / Skills                                  |
|-------------------|-------------------------------------------------|
| Operating systems | Linux (Ubuntu, RHEL), Windows Server basics     |
| Shell             | Bash, Zsh, PowerShell basics                    |
| Text editing      | Vim, Nano, VS Code                              |
| Networking        | TCP/IP, DNS, HTTP/HTTPS, TLS, ports, firewalls  |
| Version control   | Git (clone, branch, merge, rebase, cherry-pick) |
| Scripting         | Python, Bash, Go (pick one primary)             |
| Documentation     | Markdown, diagrams (draw.io, Mermaid)           |

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## 2.2 Level 1 — DevOps Foundations

### 2.2.1 Concepts

- What is DevOps? Culture, CALMS (Culture, Automation, Lean, Measurement, Sharing)
- DevOps vs SysAdmin vs SRE vs Platform Engineering
- Continuous Integration (CI) vs Continuous Delivery (CD) vs Continuous Deployment
- Build artifacts, immutable infrastructure
- Configuration vs secrets
- The Twelve-Factor App
- DevSecOps introduction
- GitOps introduction

### 2.2.2 Tools

| Category    | Tools                                         |
|-------------|-----------------------------------------------|
| Git hosting | GitHub, GitLab, Bitbucket, Azure DevOps Repos |
| SSH & keys  | OpenSSH, ssh-agent, key pairs                 |

|                  |                                    |
|------------------|------------------------------------|
| Package managers | apt, yum/dnf, Homebrew             |
| Build tools      | Make, Maven, Gradle, npm/yarn/pnpm |
| Virtualization   | VirtualBox, VMware, Hyper-V        |
| Cloud CLI basics | AWS CLI, gcloud, az                |

## 2.3 Level 2 — Infrastructure & Automation

### 2.3.1 Concepts

- Infrastructure as Code (IaC) principles
- Idempotency, declarative vs imperative
- Server provisioning & bootstrapping
- Configuration management
- Immutable vs mutable infrastructure
- Golden images / AMIs
- Inventory & state management

### 2.3.2 Tools

| Category           | Tools                                                                                       |
|--------------------|---------------------------------------------------------------------------------------------|
| IaC — Terraform    | Terraform, OpenTofu, Terragrunt                                                             |
| IaC — Cloud-native | AWS CloudFormation, Azure Bicep/ARM, Google Deployment Manager, Pulumi, CDK (AWS/Azure/GCP) |
| Configuration mgmt | Ansible, Chef, Puppet, SaltStack                                                            |
| Image building     | Packer                                                                                      |
| Secrets (intro)    | HashiCorp Vault, AWS Secrets Manager, SOPS                                                  |
| Cloud VMs          | EC2, Azure VMs, GCE                                                                         |

## 2.4 Level 3 — Containers & Orchestration

### 2.4.1 Concepts

- Containers vs VMs
- Images, layers, registries
- Dockerfile best practices
- Container networking & storage
- Orchestration primitives (Pod, Service, Deployment)
- Service mesh concepts
- Sidecar pattern
- Helm charts & package management

### 2.4.2 Tools

| Category        | Tools                             |
|-----------------|-----------------------------------|
| Runtime         | Docker, containerd, CRI-O, Podman |
| Compose / local | Docker Compose, Dev Containers    |

|               |                                                    |
|---------------|----------------------------------------------------|
| Registries    | Docker Hub, ECR, GCR, ACR, Harbor, Artifactory     |
| Orchestration | Kubernetes (K8s), Amazon ECS, Amazon EKS, GKE, AKS |
| K8s tooling   | kubectl, k9s, Lens, Helm, Kustomize                |
| Service mesh  | Istio, Linkerd, Consul Connect                     |
| Local K8s     | minikube, kind, k3s, Docker Desktop K8s            |

## 2.5 Level 4 — CI/CD Pipelines

### 2.5.1 Concepts

- Pipeline stages: source → build → test → scan → deploy
- Branching strategies (GitFlow, trunk-based)
- Artifact versioning & promotion
- Blue/green, canary, rolling deployments
- Feature flags
- Pipeline as Code
- DORA metrics (deployment frequency, lead time, MTTR, change failure rate)

### 2.5.2 Tools

| Category        | Tools                                                                                                                |
|-----------------|----------------------------------------------------------------------------------------------------------------------|
| CI/CD platforms | Jenkins, GitHub Actions, GitLab CI, CircleCI, Travis CI, Azure Pipelines, AWS CodePipeline, Tekton, Argo CD (GitOps) |
| Build agents    | Jenkins agents, GitHub-hosted/self-hosted runners                                                                    |
| Artifact repos  | Nexus, Artifactory, GitHub Packages                                                                                  |
| GitOps          | Argo CD, Flux, Fleet                                                                                                 |
| Deployment      | Spinnaker, Harness, Octopus Deploy                                                                                   |

## 2.6 Level 5 — Cloud Platforms (cross-cutting)

### 2.6.1 Concepts

- Shared responsibility model
- Regions, AZs, edge locations
- IAM (users, roles, policies, least privilege)
- VPC networking (subnets, routing, NAT, peering)
- Load balancing (L4 vs L7)
- Object storage vs block vs file
- Managed vs self-managed services
- Well-Architected Framework pillars
- Cost optimization & tagging
- Multi-account / landing zone strategy

### 2.6.2 Tools (by provider)

| Provider | Core services to learn |
|----------|------------------------|
|----------|------------------------|

|       |                                               |
|-------|-----------------------------------------------|
| AWS   | See <a href="#">curriculum-aws.md</a>         |
| Azure | Resource Manager, AAD, VNet, AKS, App Service |
| GCP   | IAM, VPC, GKE, Cloud Run, Cloud Build         |

## 2.7 Level 6 — Observability & Reliability

### 2.7.1 Concepts

- Monitoring vs observability (metrics, logs, traces)
- SLI, SLO, SLA, error budgets
- Alerting philosophy (symptom vs cause)
- On-call, incident management, postmortems
- Chaos engineering
- Capacity planning
- Distributed tracing

### 2.7.2 Tools

| Category      | Tools                                                                            |
|---------------|----------------------------------------------------------------------------------|
| Metrics       | Prometheus, Grafana, Datadog, New Relic, CloudWatch, Azure Monitor               |
| Logging       | ELK/EFK (Elasticsearch, Logstash/Fluentd, Kibana), Loki, Splunk, CloudWatch Logs |
| Tracing       | Jaeger, Zipkin, OpenTelemetry, AWS X-Ray, Honeycomb                              |
| APM           | Datadog APM, Dynatrace, AppDynamics                                              |
| Incident mgmt | PagerDuty, Opsgenie, incident.io                                                 |
| Status        | Statuspage                                                                       |
| Chaos         | Chaos Monkey, LitmusChaos, Gremlin                                               |

## 2.8 Level 7 — Security (DevSecOps)

### 2.8.1 Concepts

- Shift-left security
- OWASP Top 10
- Supply chain security (SBOM, signing)
- Zero trust networking
- CIS benchmarks
- Compliance frameworks (SOC 2, PCI-DSS, HIPAA overview)
- Threat modeling
- Principle of least privilege

### 2.8.2 Tools

| Category | Tools                                 |
|----------|---------------------------------------|
| SAST     | SonarQube, Semgrep, CodeQL, Checkmarx |

|                |                                            |
|----------------|--------------------------------------------|
| DAST           | OWASP ZAP, Burp Suite                      |
| Container scan | Trivy, Gype, Snyk, Clair                   |
| IaC scan       | Checkov, tfsec, KICS, cfn-lint             |
| Secrets scan   | git-secrets, TruffleHog, Gitleaks          |
| Policy as code | OPA, Gatekeeper, Kyverno, Sentinel         |
| WAF / edge     | AWS WAF, Cloudflare, ModSecurity           |
| PKI / certs    | cert-manager, Let's Encrypt, ACM           |
| SIEM           | Splunk, Elastic Security, AWS Security Hub |

## 2.9 Level 8 — Data, Messaging & Databases (Ops view)

### 2.9.1 Concepts

- Backup & restore strategies (RPO/RTO)
- Replication, failover, DR
- Connection pooling
- Caching layers
- Message queues & event-driven architecture
- Database migrations in CI/CD

### 2.9.2 Tools

| Category   | Tools                                                 |
|------------|-------------------------------------------------------|
| RDBMS      | PostgreSQL, MySQL, MariaDB, SQL Server                |
| NoSQL      | MongoDB, DynamoDB, Redis, Cassandra                   |
| Messaging  | Kafka, RabbitMQ, SQS, SNS, Pub/Sub                    |
| Streaming  | Kinesis, Flink, Spark Streaming                       |
| Backup     | Velero (K8s), AWS Backup, pg_dump, Percona XtraBackup |
| Migrations | Flyway, Liquibase, Alembic                            |

## 2.10 Level 9 — Platform Engineering & Internal Developer Platforms (adv)

### 2.10.1 Concepts

- Platform as a product
- Developer experience (DevEx)
- Self-service portals
- Golden paths & paved roads
- Service catalogs
- Backstage software templates

### 2.10.2 Tools

| Category | Tools |
|----------|-------|
|          |       |

|                   |                                           |
|-------------------|-------------------------------------------|
| IDP               | Backstage, Port, Cortex                   |
| API gateways      | Kong, Apigee, AWS API Gateway, Ambassador |
| Service discovery | Consul, Eureka, CoreDNS                   |
| Feature flags     | LaunchDarkly, Unleash, Flagsmith          |

## 2.11 Level 10 — Advanced & Specialized (adv)

### 2.11.1 Concepts

- Site Reliability Engineering (SRE) — toil reduction, automation
- FinOps & cloud cost governance
- Multi-cloud & hybrid cloud
- Edge computing & CDN strategy
- WebAssembly in ops (adv)
- eBPF for observability (adv)
- ML/LLM ops (MLOps) pipelines

### 2.11.2 Tools

| Category | Tools                                               |
|----------|-----------------------------------------------------|
| FinOps   | Kubecost, CloudHealth, AWS Cost Explorer, Infracost |
| CDN      | CloudFront, Cloudflare, Fastly, Akamai              |
| DNS      | Route 53, Cloudflare DNS, ExternalDNS               |
| Workflow | Airflow, Temporal, Step Functions                   |
| MLOps    | MLflow, Kubeflow, SageMaker (AWS)                   |
| eBPF     | Cilium, Pixie, Falco                                |

## 2.12 Certifications (optional milestones)

| Track      | Certifications                                                             |
|------------|----------------------------------------------------------------------------|
| Linux      | LPIC, RHCSA/RHCE                                                           |
| Cloud      | AWS SAA, AWS DVA, AWS SOA, AWS DOP, AWS SAP; Azure AZ-104, AZ-400; GCP PCA |
| Kubernetes | CKA, CKAD, CKS                                                             |
| Terraform  | HashiCorp Terraform Associate                                              |
| Security   | CISSP (broad), AWS Security Specialty                                      |
| SRE        | Google SRE book + practice (no single cert)                                |

## 2.13 Learning project ideas (apply everything)

1. **Static site pipeline** — S3 + CloudFront + GitHub Actions + Terraform

2. **Microservices on EKS** — Helm, Prometheus, Argo CD
3. **Full observability stack** — app + OpenTelemetry + Grafana stack
4. **Secure CI/CD** — SAST/DAST/SCA in pipeline, signed images
5. **Multi-env IaC** — dev/stage/prod with Terragrunt or Terraform workspaces
6. **Incident simulation** — chaos tests + runbooks + PagerDuty
7. **Internal developer portal** — Backstage with one golden-path service